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Pattern Matching and Machine Learning for Audio Signal Processing
Exercise sheet 8

To be uploaded in eCampus till: 05-07-2026 22:00 (strict deadline)

Exercise 8.1

[$2 + 3 + 3 = 8$ points]

In this exercise you have to implement the index-based audio retrieval task which you have seen in the lecture starting from the constellation maps of the database and the query. In order to do so you have to implement the following steps:

(a) Create a function which computes the inverted lists for a given constellation map (binary matrix) of a database.

(b) Compute the indicator functions and the matching function.

(c) Provide a script that tests your audio retrieval method with the constellation maps given on slide 16 of Chapter 4.

Exercise 8.2

[$4 + 3 = 7$ points]

In this exercise you have to implement an easy example of audio identification following what you have learned in the lecture. In order to do so you have to implement the following steps:

(a) Write a function that extracts peaks of a spectrogram by following the idea given on slide 8 in Chapter 4. The output should be a binary matrix on which the function of Exercise 7.3 can be applied.

(b) Use the three audio examples given on eCampus as an audio database and search for the query provided as well on eCampus using the functions created in Exercise 8.1. Identify the audio example the query is taken from and return the time stamps of the query within the audio example.